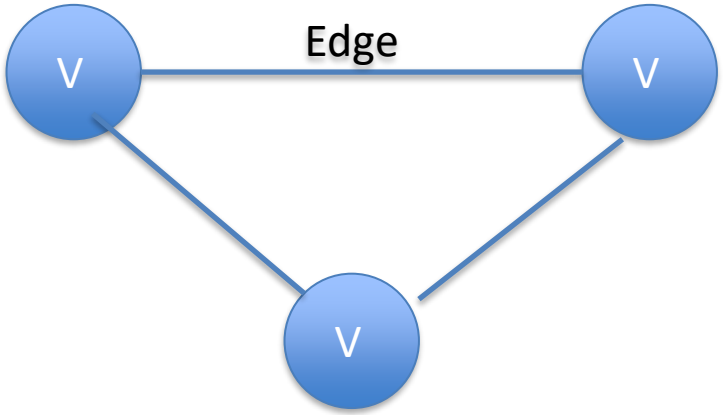
Components of a Graph

- **Vertex (Node):** represents an object
- **Edge:** connection between two nodes
- **Example:** Cities connected by roads



Directed Graph (Real-life Example)

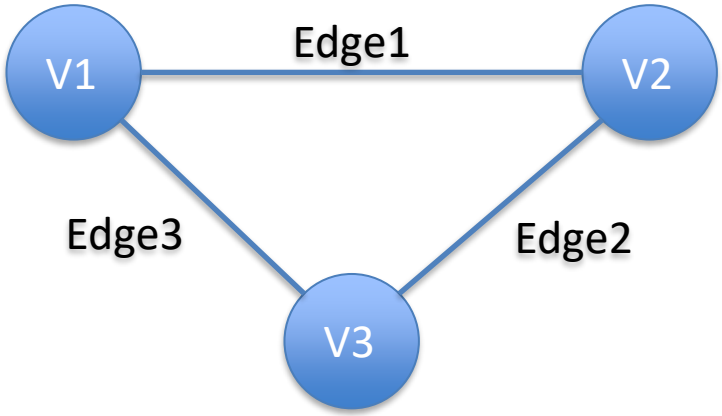
- Example: Instagram/Twitter Follow
- If A follows B, it doesn't mean B follows A

Undirected Graph (Real-life Example)

- Example: Facebook friendship
- If A is friend of B, B is friend of A

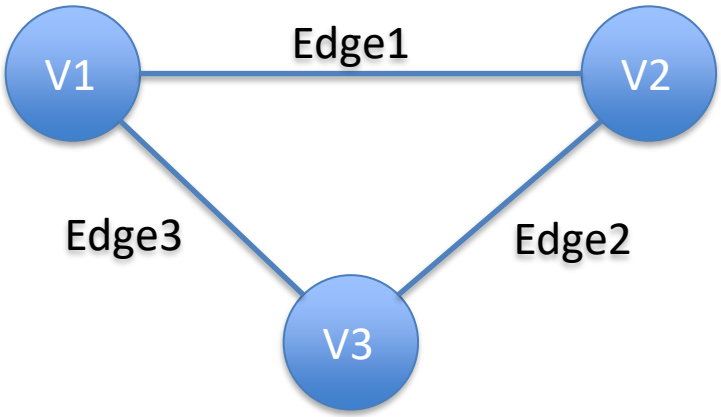
Connected Graph

- ✓ A graph is connected if every node can reach every other node
- ✓ There exists at least one path between nodes



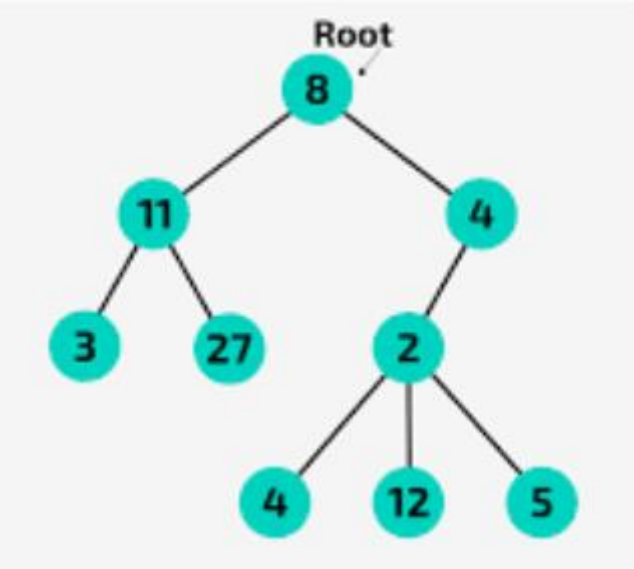
Fully Connected Graph (Complete Graph)

- Every pair of nodes has a direct edge



Is a Tree a Connected Graph?

- Yes. Trees are always connected
- Every node can reach another node
- Trees do not contain cycles

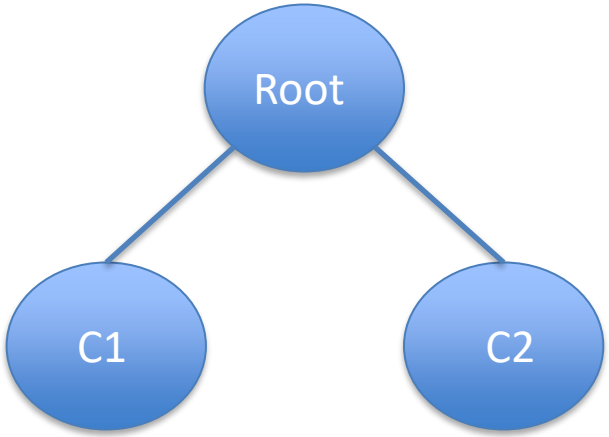


What Makes a Tree Different from a Graph?

- ✓ A Tree is a special graph:
  - Connected
  - No cycles
  - Edges = n − 1

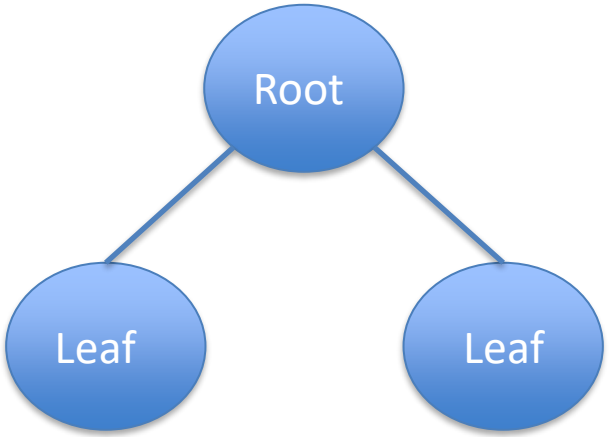
Rooted Tree

- One node is selected as the root
- Parent-child relationships exist



Leaf Node

- A node with no children
- Located at the bottom of a tree



Number of Edges in a Tree

- If number of nodes = n
- Edges = n − 1
- Example: 5 nodes → 4 edges

Simple Paths Between Two Nodes in a Tree

- There is exactly ONE simple path
- Because trees do not contain cycles

Weighted Graph

- Edges have weights (cost, distance, time)
- Example: Distance between cities